

4/4/26

NLP - Tutorial 4

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1) Cosine and Jaccard Similarity:

S1: Machine Learning is interesting.

S2: Deep Learning is Powerful.

Step - 1: Tokenisation

S1: { Machine, Learning, is, Interesting }

S2: { Deep, Learning, is, Powerful }

Step - 2: Union

All unique words,

{ Machine, Deep, Learning, is, Interesting, Powerful }

Step - 3: Vector Representation

word	S1	S2
Machine	1	0
Learning	1	1
Deep	0	1
is	1	0
Interesting	1	0
Powerful	0	1

Step-4: Cosine Similarity

$$\cos \theta = \frac{A \cdot B}{|A||B|}$$

$$A \cdot B = (1 \times 0) + (1 \times 1) + (0 \times 1) + (1 \times 1) + (1 \times 0) + (0 \times 1)$$

$$= 2$$

$$|A| = \sqrt{1+1+1+1} = \sqrt{4} = 2$$

$$|B| = \sqrt{1+1+1+1} = \sqrt{4} = 2$$

$$\cos \theta = \frac{2}{2 \times 2} = \frac{1}{2} = 0.5$$

Step-5: Jaccard Similarity

$$J = \frac{|\text{Intersection}|}{|\text{Union}|}$$

$$\text{Intersection} = \{ \text{learning, is} \} = 2$$

$$\text{Union} = \{ \text{Machine, Deep, powerful, interesting, learning, is} \} = 6$$

$$J = \frac{2}{6} = 0.33$$

2) Decentralization

a) Morphemes - De + central + ize + ation

b) Root - central

Prefix - De and Suffix - ize, -ation

- c) classification - de - Derivational prefix
 central - free morpheme
 ize - Derivational suffix
 ation - Derivational Suffix.

Unbelievable

- a) morphemes - Un + believe + able
 b) Root - Believe, Prefix - un, Suffix - able
 c) ~~Believe~~^{Un} - Derivational prefix
 Believe - Free Morpheme
 able - Derivational suffix

Replaying

- a) morphemes - Re + play + ing
 b) Root - play, prefix - Re, Suffix - ing
 c) classification - re - Derivation prefix
 play - Free Morpheme
 ing - Inflectional Suffix.

Disagreement

- a) morphemes - Dis + agree + ment
 b) Root - Agree, prefix - Dis, Suffix - ment
 c) classification - Dis - Derivational prefix
 Agree - Free morpheme

ment - Derivational suffix

Careless

- a) Morphemes - care + less
- b) Root - care, Suffix - less
- c) classification - care - free morpheme
less - Derivational suffix

4) Error Finding:

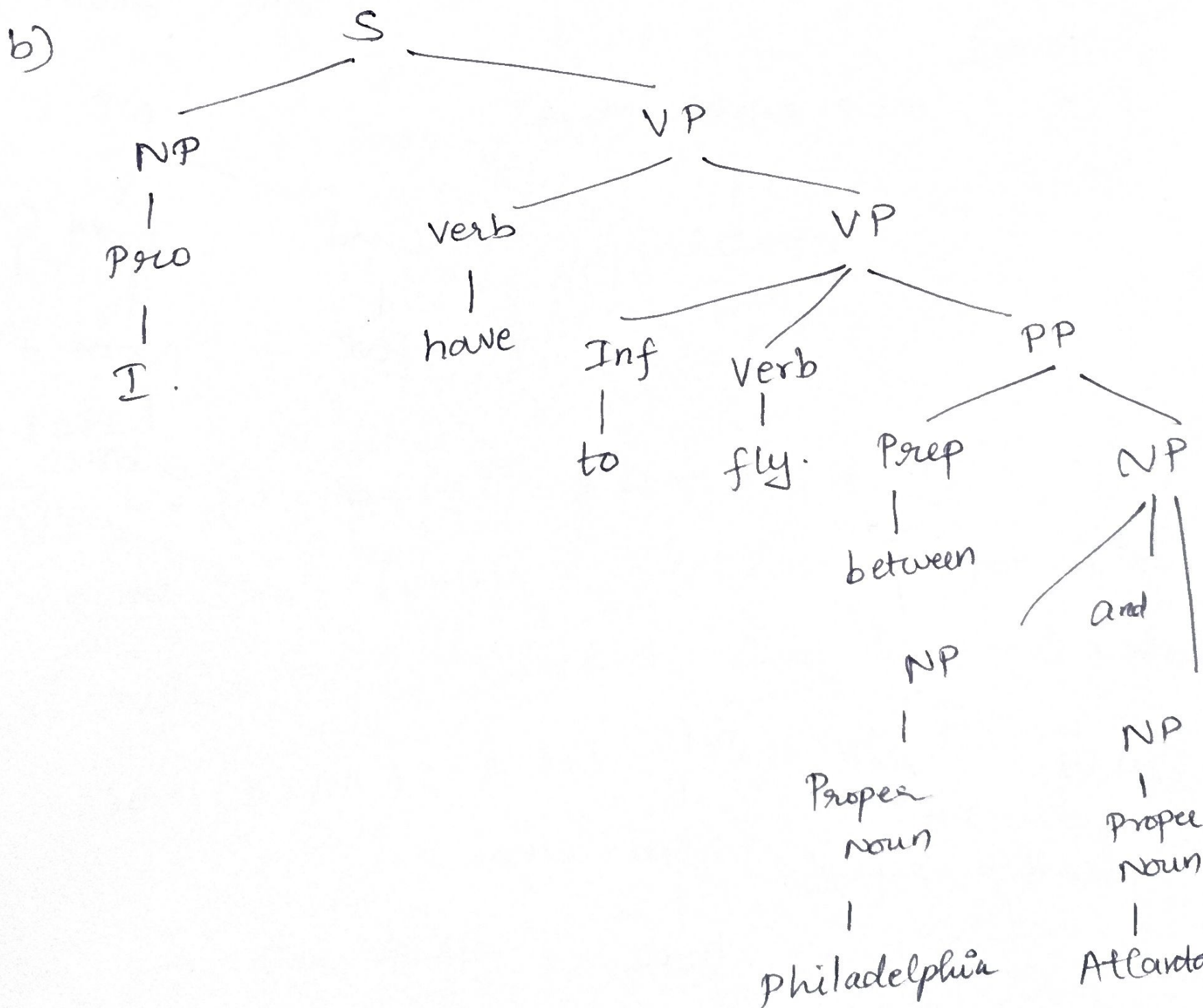
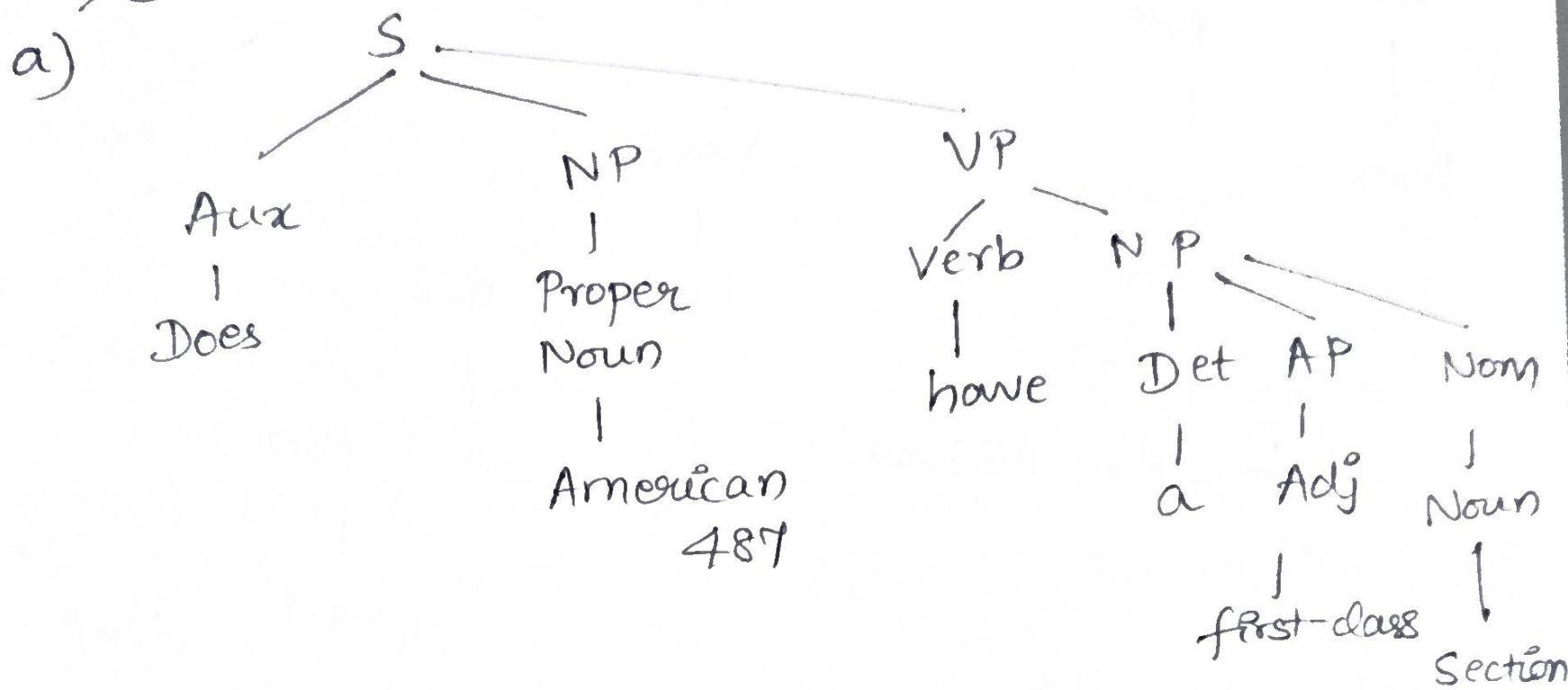
- a) Atlanta / NNP
- b) Dinner / NN
- c) have / VBP
- d) Can / MD.

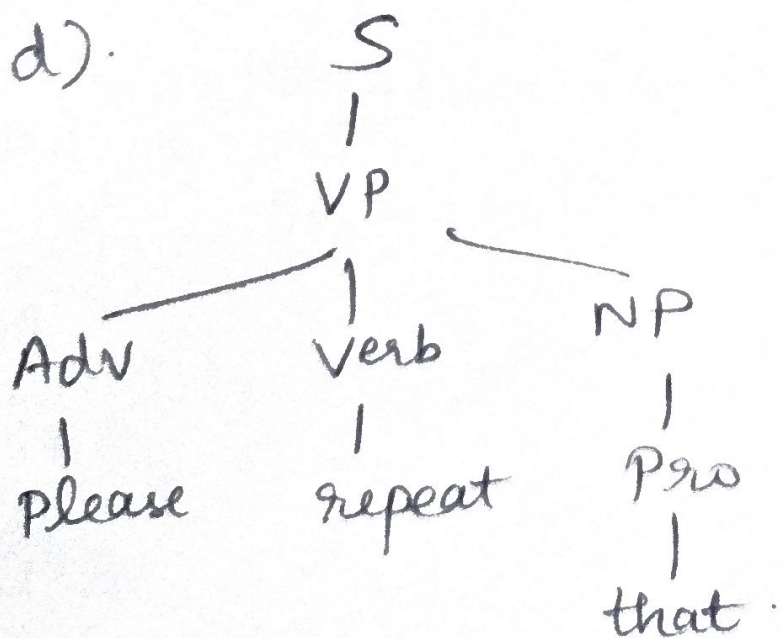
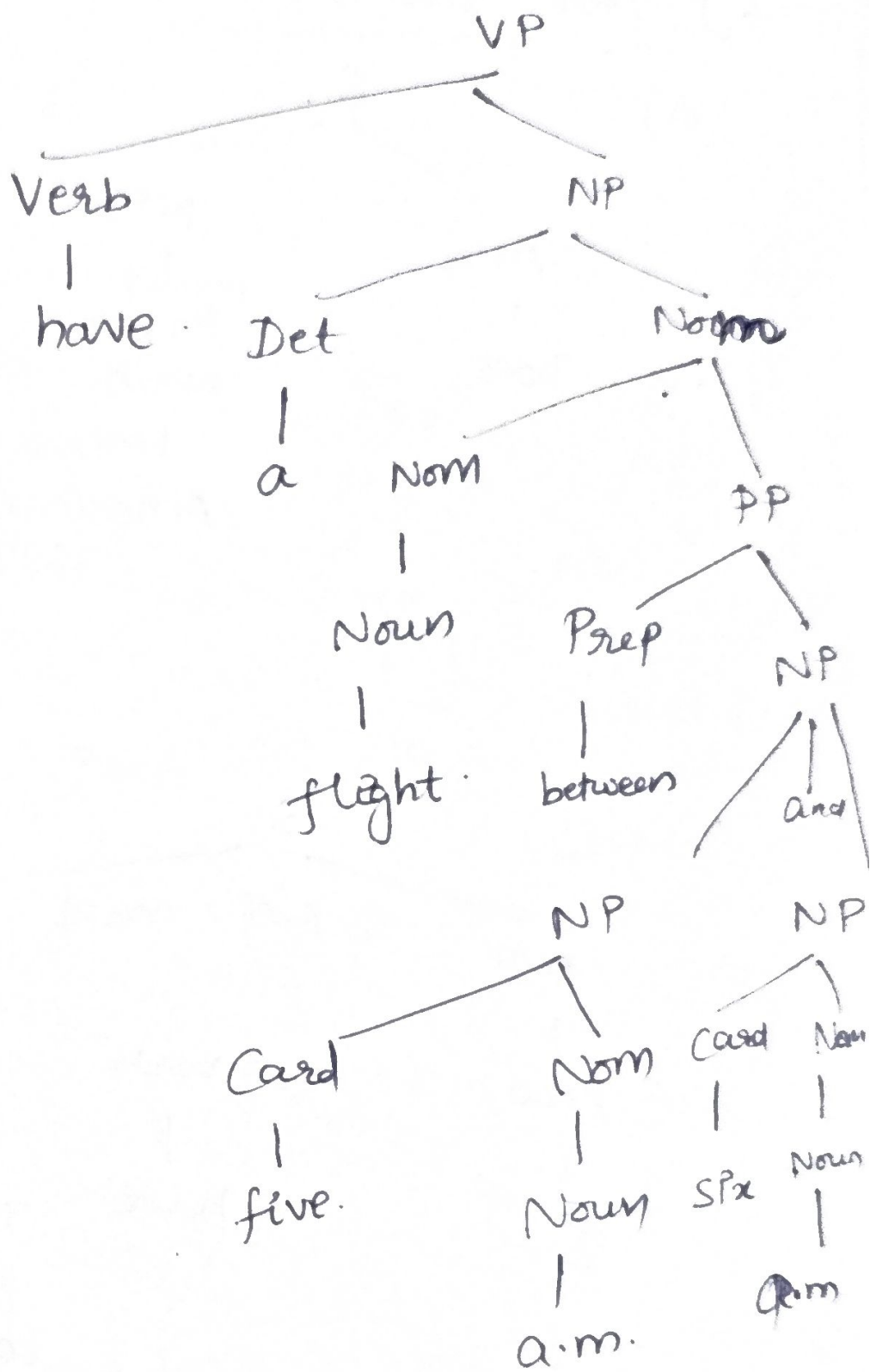
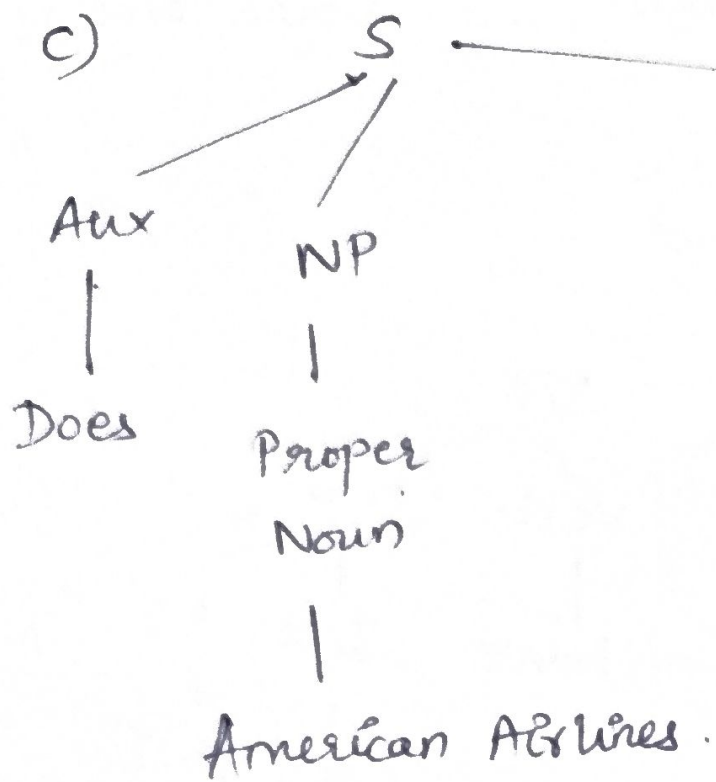
5) Penn Treebank Tagset:

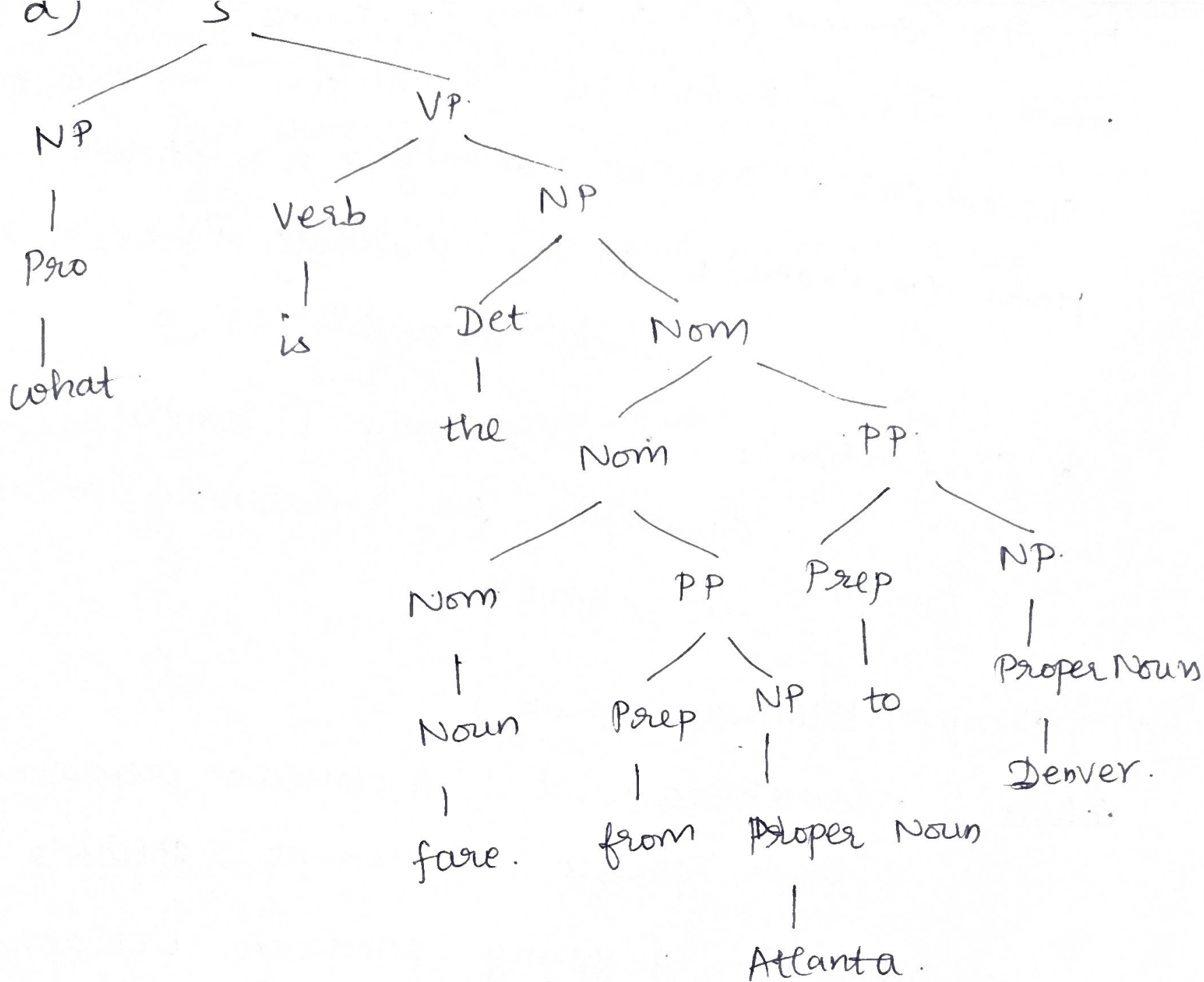
- a) It / PRP is / VBZ a / DT nice / JJ night / NN
- b) This / DT crap / NN game / NN is / VBZ
over / IN ~~Fifty~~ a / DT garage / NN in / IN
Fifty-second / NNP street / NNP...
- c) Nobody / NN ever / RB takes / VBZ the / DT
newspapers / NNS she / PRP sell / VBZ.
- d) He / PRP is / VBZ a / DT tall / JJ skinny /
JJ guy / NN with / IN a / DT long / JJ
sad / JJ mean-looking / JJ kisser / NN

and / cc a / DT mournful / JJ voice / NN .

6) Tree structures:







7) FOPC:

a) $\forall x (\text{student}(x) \Rightarrow \text{Loves}(x, AI))$

where, $\forall x$ (Universal Quantifier) — for all x

$\text{student}(x)$: Predicate indicating x is a student

$\text{Loves}(x, AI)$: A binary predicate stating x loves the object AI .

$\rightarrow$: Implication.

b) $\exists x (\text{Student}(x) \wedge \text{Hates}(x, \text{Exams}))$

where, $\exists x$ - Existential Quantifier - for some x

$\text{Student}(x)$: Predicate indicating x is a student

$\text{Hates}(x, \text{Exams})$: Binary predicate indicating x hates exams.

$\wedge$ (Conjunction): ' $\wedge$ ' with only Existential Quantifier for representing at least one object.

c) $\text{cheap}(\text{BlinksRestaurant})$

where, BlinksRestaurant : A constant representing the specific restaurant, Blink's.

$\text{cheap}(y)$ - A unary predicate indicating that an object y is cheap.

d) $\text{Delicious}(\text{DominoesPizza})$

where, DominoesPizza - A constant representing the specific entity, Domino's Pizza.

$\text{Delicious}(y)$ - A unary predicate indicating that an object y is delicious.

3) State Transition Table:

i) K - insertion

Current State	Inp: Out	Next State
q_0 (start)	Vowel: vowel	q_1
q_0	other: other	q_0
q_1	c: c	q_2
q_2	ε: k	q_3
q_1	Vowel: vowel	q_1

	ed	#	other	vowel	^:ε	c	^:k	ed
q_0	φ	0	2	2	0	φ	φ	φ
q_1	φ	φ	0	φ	φ	φ	φ	φ
q_2	φ	0	0	2	0	3	φ	φ
q_3	φ	6	0	φ	5	φ	4	φ
q_4	φ	φ	φ	φ	φ	φ	φ	6
q_5	1	0	0	φ	φ	φ	φ	φ
q_6	φ	φ	φ	φ	φ	φ	φ	φ

ii) E-deletion

	#	other	$\wedge : \mathcal{E}$	z	s	x	$\cdot \mathcal{E} : \phi$
q_0	0	0	0	1	1	1	ϕ
q_1	ϕ	ϕ	2	1	1	1	ϕ
q_2	0	0	ϕ	ϕ	5	ϕ	3
q_3	ϕ	ϕ	ϕ	ϕ	4	ϕ	ϕ
q_4	0	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ
q_5	ϕ	ϕ	2	1	1	1	ϕ

iii) consonant spelling.

	V	C	#	$\wedge : \varepsilon$	Other	Prng, ed	$\wedge : \text{same } C$
q_0	5	1	6	ϕ	ϕ	ϕ	ϕ
q_1	5	1	2	0	ϕ	ϕ	ϕ
q_2	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ
q_3	ϕ	ϕ	2	ϕ	ϕ	ϕ	ϕ
q_4	ϕ	1	2	7	ϕ	ϕ	3
q_5	5	4	6	0	ϕ	ϕ	ϕ
q_6	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ
q_7	ϕ	ϕ	0	ϕ	8	ϕ	ϕ
q_8	ϕ	ϕ	ϕ	ϕ	0	ϕ	ϕ